

Paper 1 and Paper 2: Core Pure Mathematics

To support the co-teaching of this qualification with the AS Mathematics qualification, common content has been highlighted in bold.

Topic	What students need to learn:		
	Content		Guidance
1 Proof	1.1	Construct proofs using mathematical induction. Contexts include sums of series, divisibility and powers of matrices.	To include induction proofs for (i) summation of series e.g. show $\sum_{r=1}^n r^3 = \frac{1}{4}n^2(n+1)^2$ or show $\sum_{r=1}^n r(r+1) = \frac{n(n+1)(n+2)}{3}$ (ii) divisibility e.g. show $3^{2n} + 11$ is divisible by 4 (iii) matrix products e.g. show $\begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}^n = \begin{pmatrix} 2n+1 & -4n \\ n & 1-2n \end{pmatrix}$